GITHUB DEV DAYS

Learning Effectively with AI.

A study workshop with GitHub Copilot

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WHO AM I

I'm here as a learner

Who	Mohamed Mostafa 👋. Not here as a data scientist today.
Why this	I want to really **understand** things and still remember them next month.
Today	Learn a new technique and have fun.

90 MINUTES

Agenda

Time	Activity
00–10	Intro + the Monty Hall problem
10–20	What learning is, and where AI helps
20–30	Live Copilot demo + topics and teams
30–40	Break
40–70	Learn with AI, test each other
70–90	Teach strangers + wrap-up

The Monty Hall problem

3 doors: 1 car, 2 goats. You pick Door 1.
The host, **who knows**, opens Door 3: a goat.

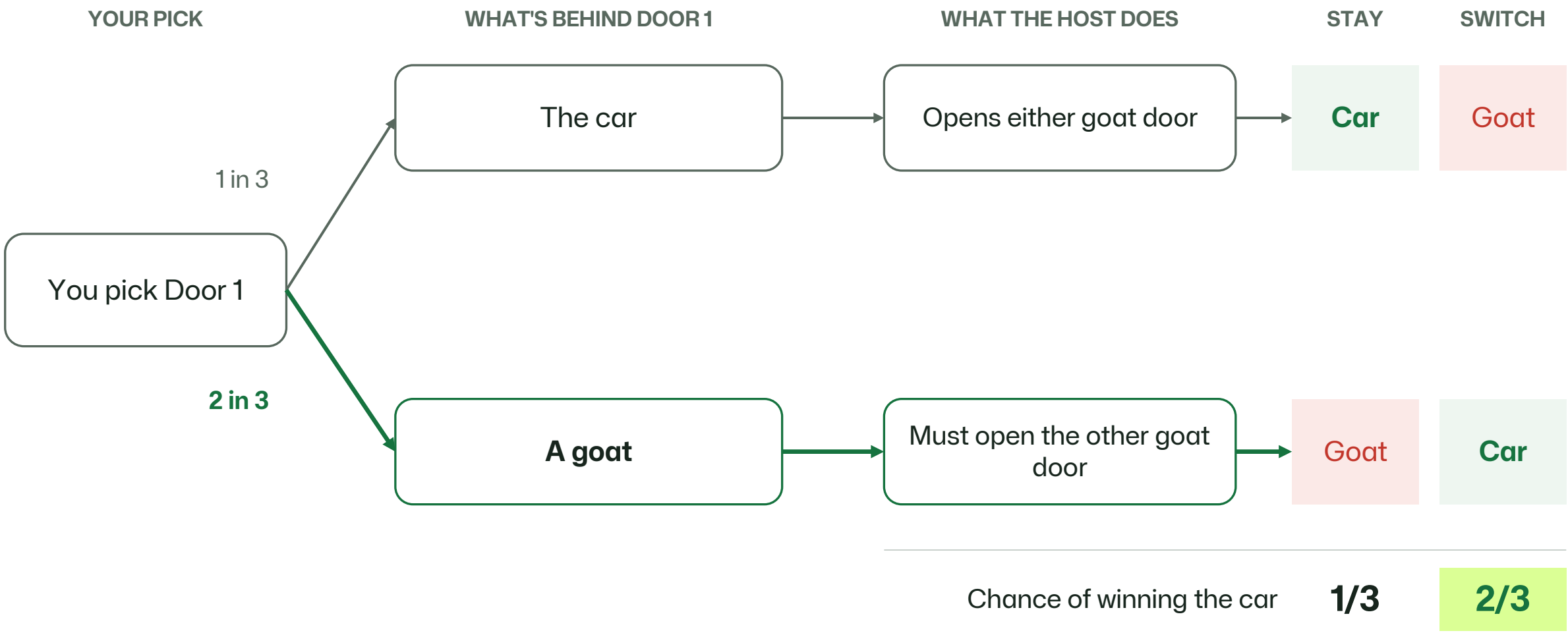
Stay with Door 1, or switch to Door 2?

Vote: stay, switch, or doesn't matter.

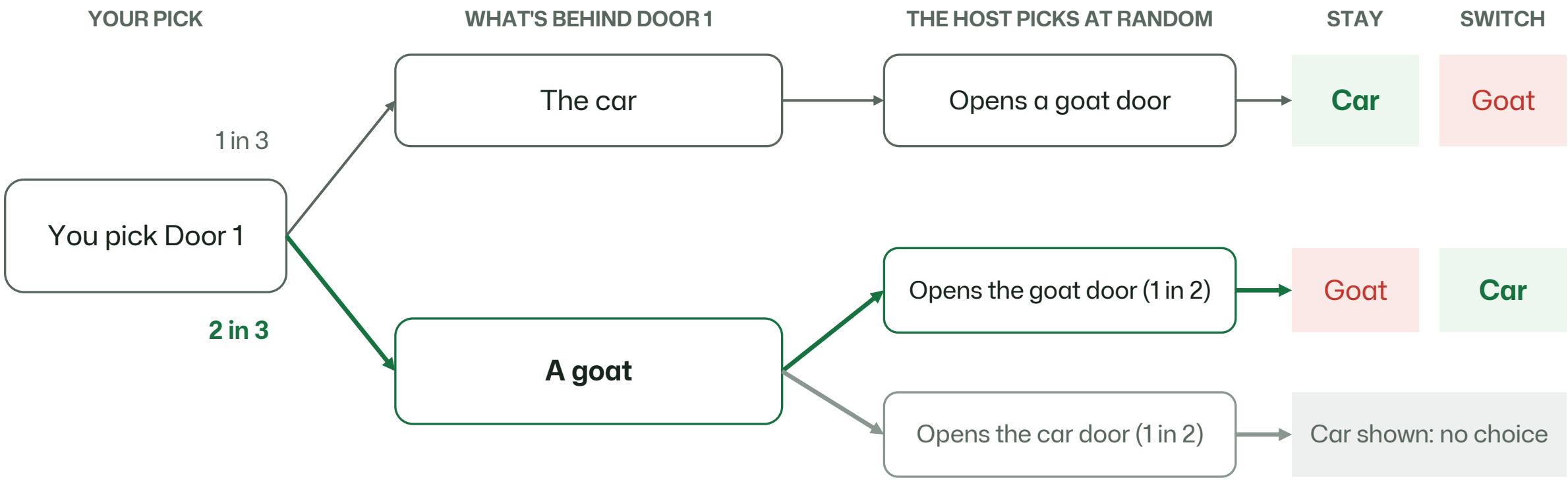
Switch wins 2 out of 3

ANSWER	Switch. You win the car 2 out of 3 times.
WHY	Your first pick is right 1 in 3. If it's wrong, the host must leave the car behind the other door.

Switch wins whenever your first pick missed



A random host makes switching a coin flip



	STAY	SWITCH
A goat is shown 1/3 of the time after a car pick, 1/3 after a goat pick. Chance of winning:	**1/2**	**1/2**

Four things real learning needs

Questions	Know what you don't know. Socratic questions expose the gaps.
Effort	Explain it and try it yourself, then get it checked.
Trust	Believe the source enough to build on it.
Retention	Still able to explain it next week, not just today.

Feeling that you understand is not the same as understanding.

Many students, many sources

Many to many	Every student uses many outlets. Every outlet serves many students.
Not your level	A lecture or a book can't start exactly where you are.
Switching cost	New style, notation and interface every time you switch.
Low trust	Your brain hedges with a source it doesn't know yet.

WHERE AI HELPS

One tutor, many sources

Fits you	Probes what you know and teaches from your edge.
Focus	Handles planning, finding and ordering. Your effort goes to the idea.
Trust	Checks claims against sources instead of asking you to just believe.
Retention	Builds flashcards and quizzes you on them in later sessions.

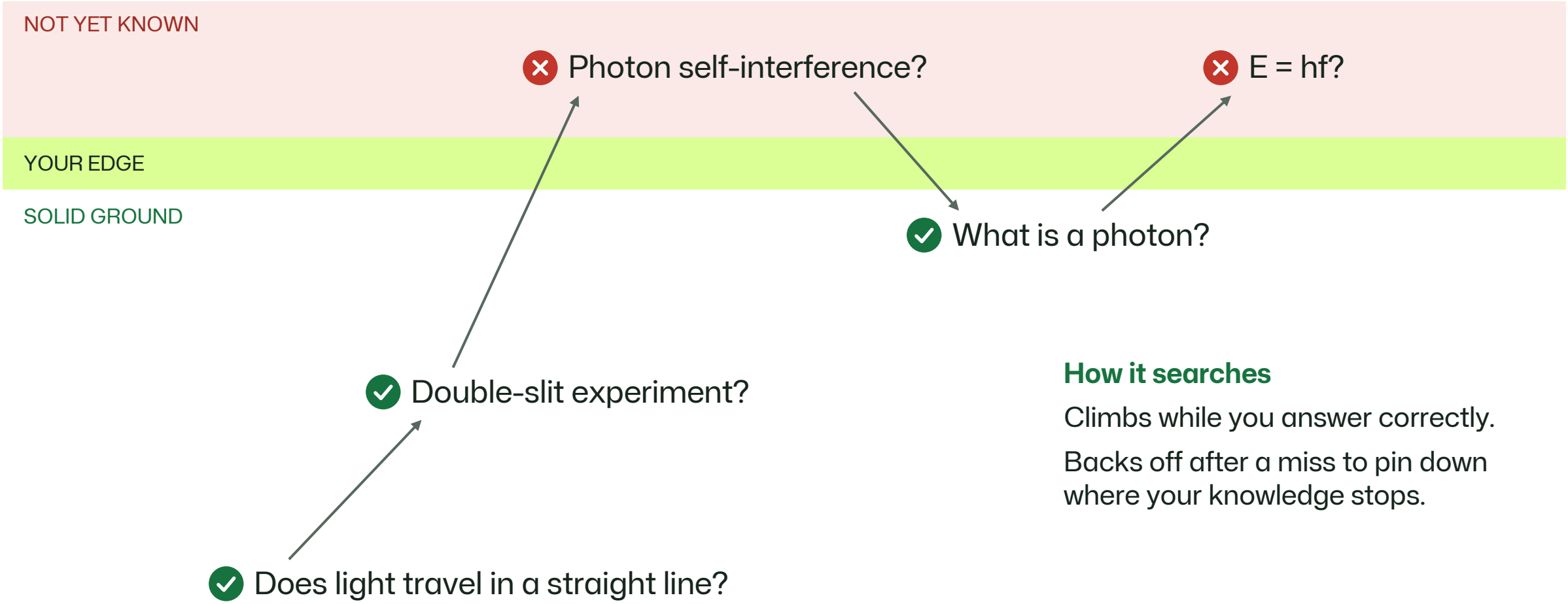
The trap: if AI hands you answers, you only feel you understand. Use it as a tutor.

LEARNING LOOP

Probe, plan, teach… and keep

Probe	The AI quizzes you to find the edge of your understanding.
Plan	A short map from what you know to your goal.
Teach	One step at a time. You try, it checks, it adapts.
Keep	Turns the lesson into flashcards.

Probe finds the edge of what you know



How it searches

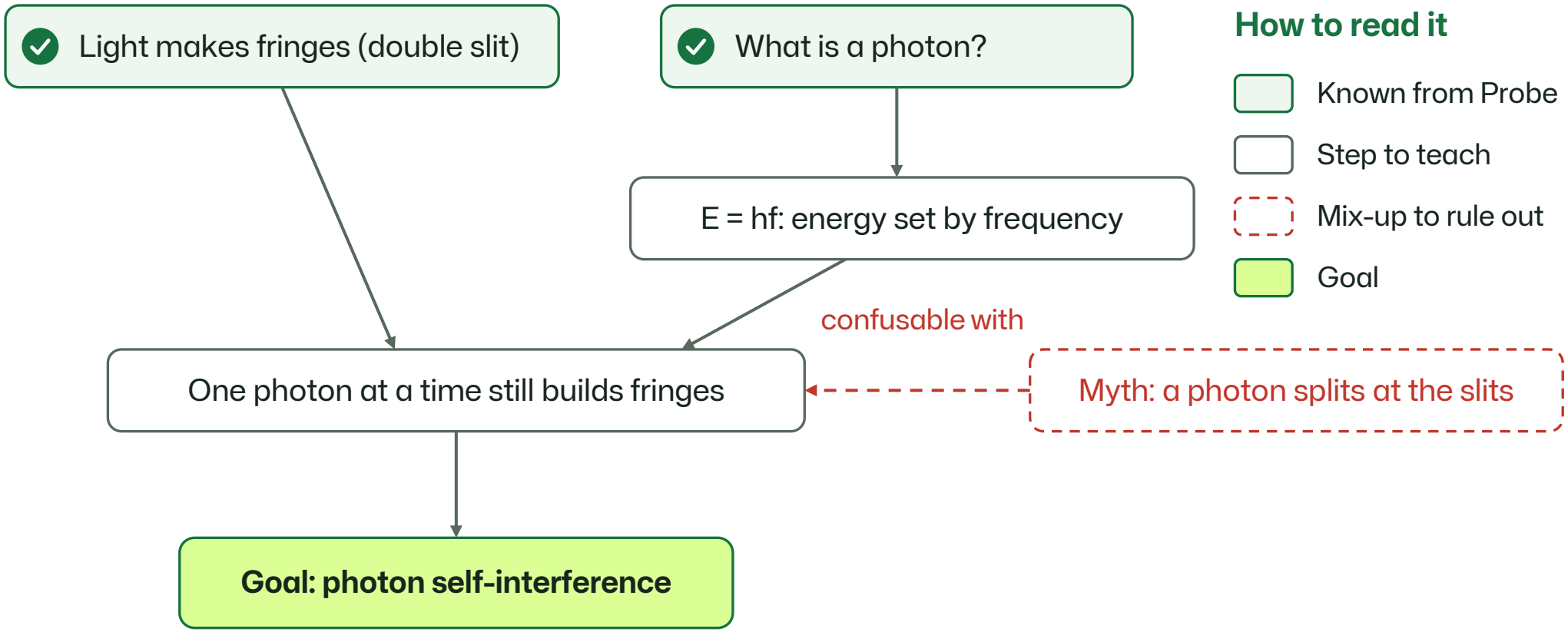
Climbs while you answer correctly.

Backs off after a miss to pin down where your knowledge stops.

Example strand

PLAN

Plan maps the steps from your edge to the goal



Each arrow means “learn this first.” Teach works through the steps top to bottom.

TEACH

A step turns Known only after you prove it

STEP FROM THE PLAN

E = hf

A photon's energy is set by its frequency, not by how bright the light is.

Marked Known when

Questions 1–3 are right, with your reasoning.

Question 4 still passes in a later session.

1 Recall	Blue light has a higher frequency than red. Which photon carries more energy?
2 Explain	Why can a dim UV lamp knock electrons off a metal when a bright red lamp can't?
3 Apply	A photon's frequency doubles. What happens to its energy, and how does E = hf show it?
4 Cold re-test	Next session, no notes: what does brightness change, and what does frequency change?

Example questions for the photon strand. A miss sends Teach back to the step before.

Keep turns each Known step into flashcards

FROM: E = hf

Which photon carries more energy, blue or red?

BACK

Blue. Its frequency is higher, so by E = hf it carries more energy.

FROM: E = hf

Why can't a bright red lamp knock electrons off a metal?

BACK

Each red photon has too little energy. Brightness adds photons, not energy per photon.

RULES OUT: THE MYTH

True or false: a photon splits in two at the double slit.

BACK

False. Each photon lands whole. Fringes build up over many photons.

Review schedule

Tomorrow → In 3 days → In 1 week → In 3 weeks

A right answer pushes the card to the next interval. A miss sends it back to tomorrow.

LIVE DEMO

Live demo: Bayes' theorem

“/teach Bayes' theorem.”

Watch for: a question, a plan, a step, a check, a flashcard deck.

Topics and teams

Topic	Teams · learning target
Algorithms	Teams 1 & 6 · Trace one binary search
Probability	Teams 2 & 7 · Explain conditional probability
Linear algebra	Teams 3 & 8 · Interpret and calculate a dot product
Databases	Teams 4 & 9 · Predict rows from a LEFT JOIN
Machine learning	Teams 5 & 10 · Calculate one gradient descent update

50 people = 10 teams of 5. Two teams per topic. Count off 1–10 to get your team.

DURING THE BREAK

Get your tutor ready

1 Open chat	VS Code → Copilot Chat. Send one test message.
2 Get the skills	github.com/MMostafa-Hub/learning-with-copilot
3 Your team	Sit at your team's table and say hi.
No Copilot?	https://www.linkedin.com/in/mohamed-mostafa-elbehery/

BREAK / 10 MINUTES

Break

Find your team table. Open Copilot Chat. Meet your teammates.

YOUR TEAM'S MISSION

Everyone can teach it alone

After this	Each of you joins a new group and teaches your topic alone.
So	Help each other. Nobody leaves this table unable to explain it.

Your team succeeds only when every member can teach it.

Your 30 minutes

Minutes	Phase	Output
0–3	Goal	One sentence the whole team agrees on
3–13	AI tutor	Each person probes, plans, starts learning
13–16	Gap huddle	Everyone names one thing they didn't know
16–23	Learn + help	Whoever gets a step explains it to whoever's stuck
23–27	Stump round	Laptops closed. Ask and answer “why?”
27–30	Prepare	A 2-minute teach-back each of you can give

JIGSAW / 16 MINUTES

One expert per topic

Number yourselves 1–5 inside your team.

Teams 1–5	All the 1s meet, all the 2s meet… one expert from every topic.
Teams 6–10	Same again. 10 new groups of 5.

Bring your brain, not your laptop.

EACH EXPERT'S TURN / 3 MINUTES

Teach, then get grilled

2 min Teach	Explain your concept with one example.
1 min Grill	Everyone else plays the AI tutor: ask “why?” and “what if?”
Then	Next expert. Five turns.

REFLECT / 2 MINUTES

Your exit ticket

One thing I can explain now

One gap I still need to work on

One person I would like to study with again

TAKE IT HOME

Your next study session

Learn	Install the skills. Let Copilot probe, plan and teach.
Keep	Turn each known idea into flashcards.
Test	Explain it to a friend. Let them ask why.

BEFORE YOU LEAVE

One small reason to meet again

“Want to study [topic] together for 30 minutes next week?”

Your learning skills repo

github.com/MMostafa-Hub/learning-with-copilot

Copy the folders inside skills/ to ~/.copilot/skills/

teach runs the loop. probe diagnoses. Keep makes flashcards.

learn-anki needs Anki and an Anki MCP server. Rehearse on a fresh account.

BACKUP / IF ASKED

Research behind the learning loop

Workshop habit	Purpose	Reference
Probe + feedback	Use answers to identify gaps and offer specific feedback.	Shute (2008)
Plan + guidance	Give novices guidance through manageable steps.	Sweller (1988)
Explain + apply	Explain why an example works in your own words.	Chi et al. (1989)
Check + revisit	Retrieve without hints, then check again another day.	Roediger & Karpicke (2006)

These findings support components of the method. The full workflow needs evaluation.

BACKUP / IF ASKED

What AI tutoring studies show

Promising result	A structured AI tutor improved immediate learning in a study of 194 undergraduate physics students. Kestin et al. (2025)
Reason for checks	An unrestricted AI assistant improved practice scores but reduced later performance without AI in high-school mathematics. Bastani et al. (2025)

Our choice: attempts before answers, checked sources, and later tests without AI.

BACKUP / IF ASKED

Research behind Probe, Teach and Keep

Illusion	Slide 7. People think they understand how things work far better than they do. Trying to explain exposes the gaps. Rozenblit & Keil, 2002
Pretesting	Slide 11. Testing before teaching helps learning, even when the answers are wrong, if the correct answer follows. Richland, Kornell & Kao, 2009
Re-testing	Slides 7 and 13. Students who practiced recall kept 61% of a passage after a week; rereaders kept 40%. Roediger & Karpicke, 2006
Spacing	Slide 14. Spread-out review beats cramming across 317 experiments. Expanding and even gaps work about the same. Cepeda et al., 2006; Latimier et al., 2021

HIDDEN APPENDIX

Sources and credits

Method	Eero Alvar: How I Use AI to Learn Things
Skill source	github.com/MMostafa-Hub/learning-with-copilot
Copilot support	GitHub Docs: Agent skills and Copilot setup
Design	github.com/github/dev-days

Research references: learning design

Feedback	Shute, V. J. (2008). Focus on Formative Feedback. Review of Educational Research, 78(1), 153–189. doi.org/10.3102/0034654307313795
Guidance	Sweller, J. (1988). Cognitive Load During Problem Solving: Effects on Learning. Cognitive Science, 12(2), 257–285. doi.org/10.1207/s15516709cog1202_4
Explanation	Chi, M. T. H., et al. (1989). Self-Explanations: How Students Study and Use Examples in Learning to Solve Problems. Cognitive Science, 13(2), 145–182. doi.org/10.1207/s15516709cog1302_1

Research references: testing and AI

Retrieval	Roediger, H. L., & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. Psychological Science, 17(3), 249–255. doi.org/10.1111/j.1467-9280.2006.01693.x
AI tutor	Kestin, G., et al. (2025). AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting. Scientific Reports, 15, 17458. doi.org/10.1038/s41598-025-97652-6
AI guardrails	Bastani, H., et al. (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. PNAS, 122(26), e2422633122. doi.org/10.1073/pnas.2422633122

Research references: probe, teach and keep

Illusion

Rozenblit, L., & Keil, F. (2002). The misunderstood limits of folk science: An illusion of explanatory depth. Cognitive Science, 26(5), 521–562. doi.org/10.1207/s15516709cog2605_1

Pretesting

Richland, L. E., Kornell, N., & Kao, L. S. (2009). The pretesting effect: Do unsuccessful retrieval attempts enhance learning? JEP: Applied, 15(3), 243–257. doi.org/10.1037/a0016496

Spacing

Cepeda, N. J., et al. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. Psychological Bulletin, 132(3), 354–380. doi.org/10.1037/0033-2909.132.3.354

Schedules

Latimier, A., Peyre, H., & Ramus, F. (2021). A meta-analytic review of the benefit of spacing out retrieval practice episodes on retention. Educational Psychology Review, 33, 959–987. doi.org/10.1007/s10648-020-09572-8

Research references: learning with others

Teaching	Fiorella, L., & Mayer, R. E. (2013). The relative benefits of learning by teaching and teaching expectancy. Contemporary Educational Psychology, 38(4), 281–288. doi.org/10.1016/j.cedpsych.2013.06.001
Peer learning	Bengesai, A. V., Amusa, L. B., & Dhunpath, R. (2023). A meta-analysis on the effect of formal peer learning approaches on course performance in higher education. Cogent Education, 10(1), 2203990. doi.org/10.1080/2331186X.2023.2203990